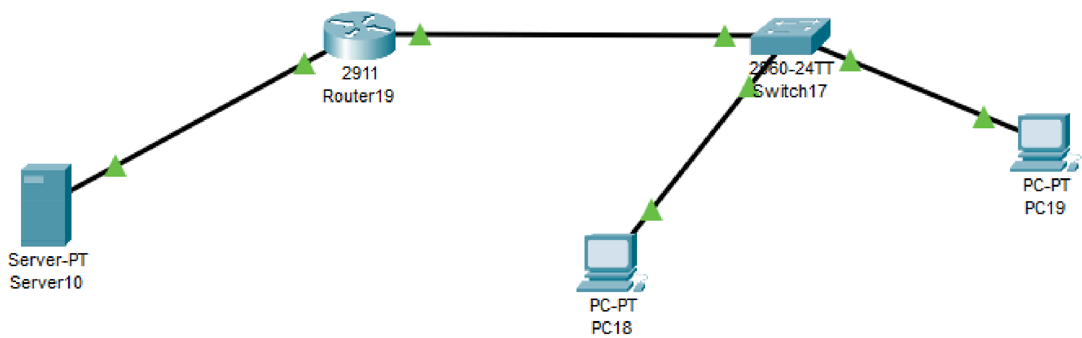
<u>DHCP</u>

This is the main project structure.
PC1, PC2 -> Switch1 -> Router19 <- Server10 (DHCP server)



I implemented a small DHCP server setup, which was moderately challenging but very insightful.

First, I connected the DHCP server to the router and configured the necessary DHCP settings. This included defining the address pool for the client PCs located on a different network segment.

DHCP Configuration:

I created a dedicated IP address pool for the client machines on the second network to ensure proper dynamic IP allocation.

DHCP settings:

IPv4 Address	192.168.10.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	0.0.0.0

Also setup a pool for the PCs on the other Network.

DHCP

Interface	FastEthernet0
Service	On
Pool Name	serverPool
Default Gateway	192.168.20.1
DNS Server	0.0.0.0
Start IP Address :	192 . 168 . 20 . 10
Subnet Mask:	255 . 255 . 255 . 0
Maximum Number of Users :	246
TFTP Server:	0.0.0.0
WLC Address:	0.0.0.0

Add | Save | Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168....	0.0.0.0	192.168....	255.255....	246	0.0.0.0	0.0.0.0

Router Configuration (ROUTER19):

The router was configured as follows:

```
en
conf t
int g0/0

ip address 192.168.10.1 255.255.255.0
no shutdown

int g0/1
ip address 192.168.20.1 255.255.255.0
ip helper-address 192.168.10.2
no shutdown
```

The ip helper-address command was used on the second interface to forward DHCP requests to the DHCP server located in the 192.168.10.0/24 network.

Final Setup:

After completing the router configuration, I connected the switch along with PCs (PC18 and PC19). Once the network setup was complete, I enabled DHCP on the client machines. The devices successfully received IP addresses from the DHCP server, confirming that the configuration was working correctly.